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Properties of Materials
Lecture 3

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Lecture 1: Module Overview, Module Aim, Why Study Materials, The World of Materials (1), History - The Evolution of Materials, Materials Selection in Design (1), Properties of Engineering Materials (1), Products, Three-Way Relationship (1), Origins of Material Properties, How Do We - As Engineers - Select Materials
Lecture 2: Concepts and Terminology (1), Properties of Engineering Materials (2), Mechanical Properties (1), Elastic Properties, Elastic Properties: Covalent and Metallic Bonds - Ionic Model - The Origin of Modulus, Plastic Properties, Fracture / Fatigue / Creep, Thermal Properties (1), Density and Corrosion, Corrosion: Anodic and Cathodic Reactions, Ellingham Diagram
Lecture 3: Structure-Property-Processing Relationship, Structure Sensitive Properties, Structure Insensitive Properties, The Three-Part Relationship, Materials Property Web Sites, Materials Selection, Material Property - Charts: Modulus - Density, Mechanical Properties (2), Thermal Properties (2), A TOOL for Materials Selection, Structure Sensitivity, Structure Sensitivity in Metals, Controlling Structure, Case Study

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Engineering Materials and Their Properties

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Properties of Engineering Materials

Mechanical
elastic
plastic
fracture/fatigue
creep

Chemical
corrosion
oxidation

Functional Materials

Physical
optical
magnetic

Thermal
thermal expansion
thermal conductivity
heat capacity
melting points
transition temperature

Environmental
recycling
energy consumption

Other
feel
look

How do we design materials to get the most out of them?

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Structure-Property-Processing Relationship

a. Structure sensitive properties

Properties sensitive to microstructures of the materials (not only the composition, but also the microstructures and how the materials have been processed)

Strength, ductility, fracture toughness, fatigue + creep properties, corrosion + wear resistance

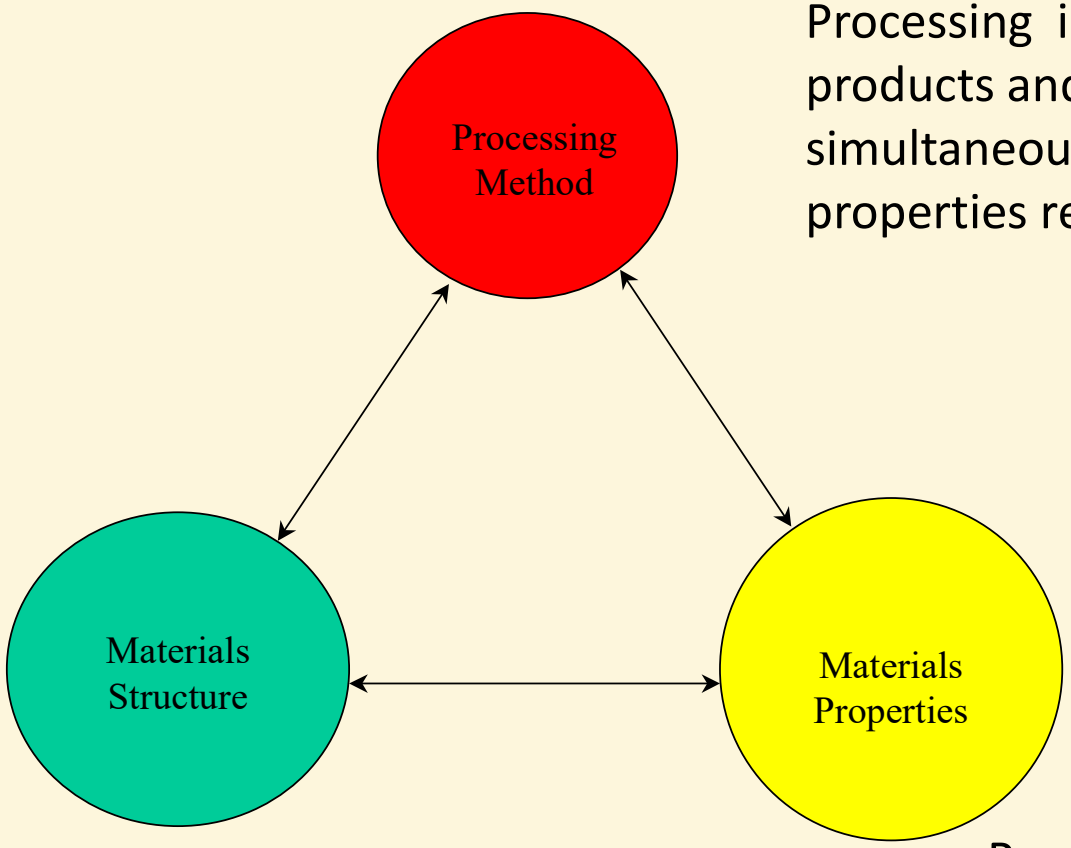
b. Structure insensitive properties

Properties insensitive (<5%) to microstructures of the materials:

Elastic properties, thermal expansion coefficient, specific heat

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The three-part relationship between structure, properties and processing



Processing involves shaping engineering products and devices, while simultaneously providing the materials properties required by the design

Understand how processing history controls microstructure and properties

Materials structure is controlled by the composition and how it is processed

Properties of materials reflect their microstructure

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Materials Property web sites

MatWeb: http://www.matweb.com/index.asp

Goodfellow: http://www.goodfellow.com/csp/active/gfMaterials.csp

Doitpoms: http://www.doitpoms.ac.uk/tlplib/index.php

Matter: http://www.matter.org.uk/universities.htm

I cannot begin to tell you how good these last two in particular are!

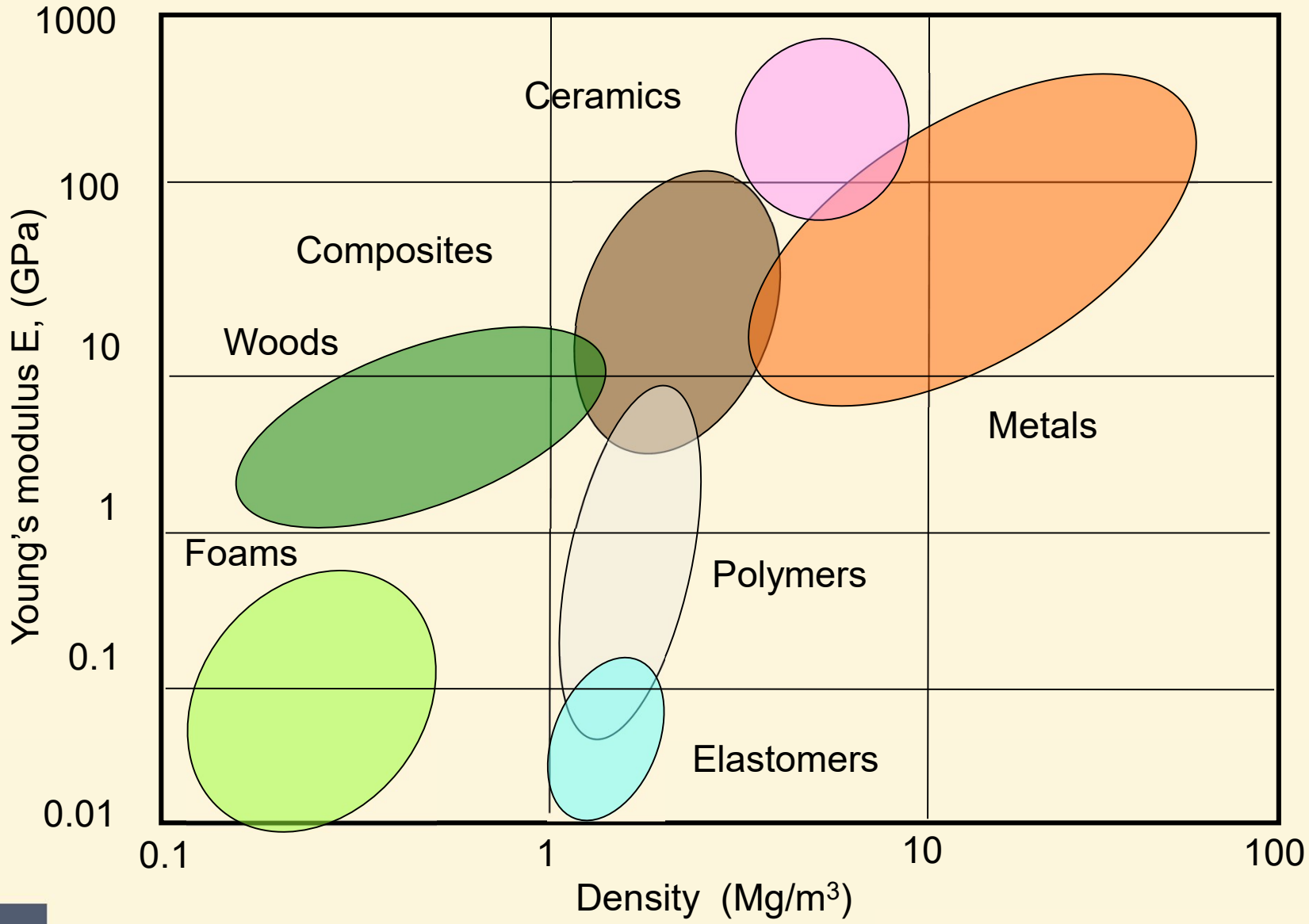
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Materials Selection

- There are hundreds of thousands of different materials
- Far too many for anyone to remember!
- So how do we choose the best material for a particular design?
- Ashby diagrams

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Material property- charts: modulus - density



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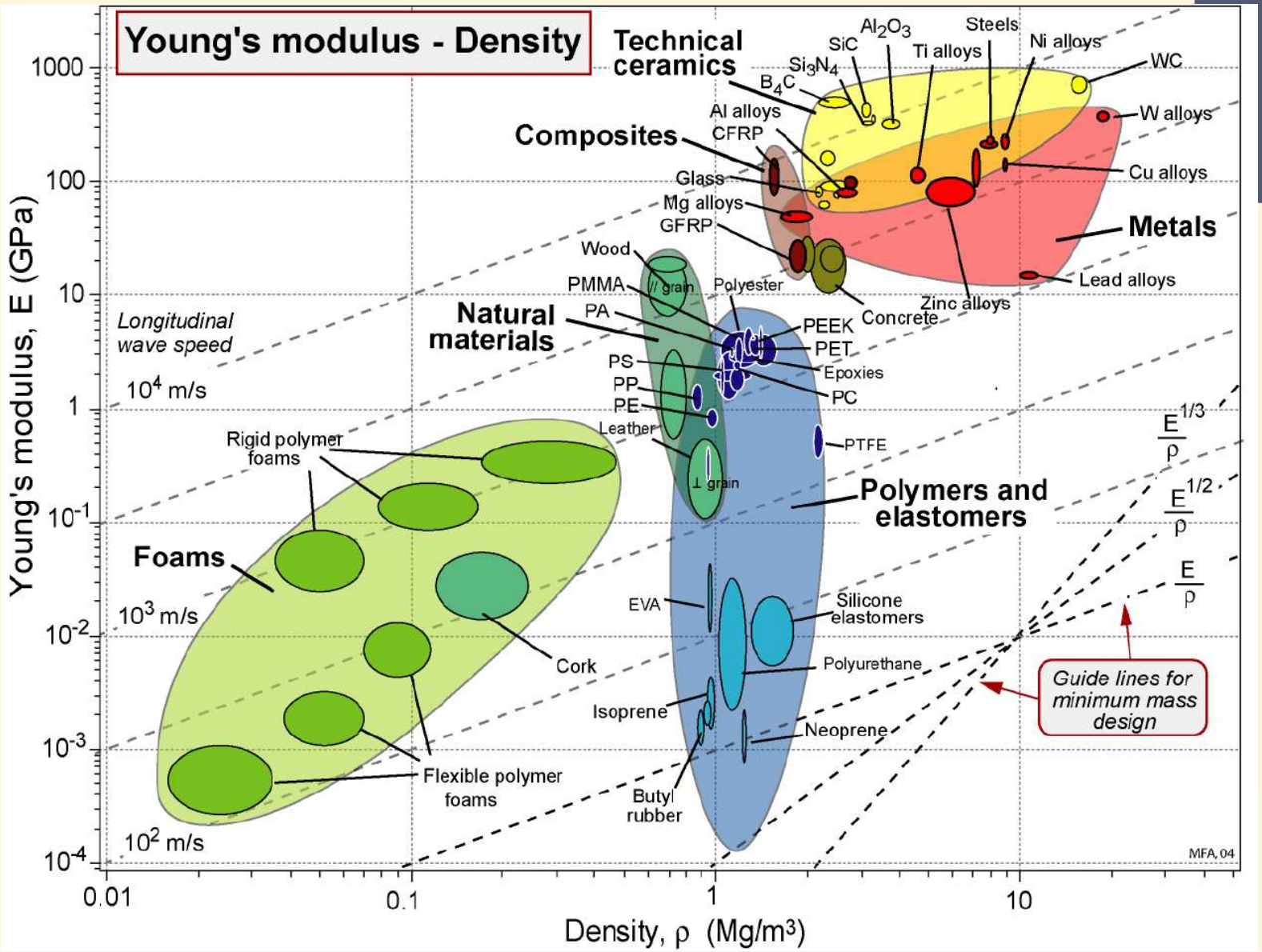
Mechanical properties

Why the differences?

- ***Atom size and weight***
- ***Bonds as (linear) springs***
- ***Spring constant for various bond types.***

Manipulating properties

- ***Making composites***
- ***Making foams***



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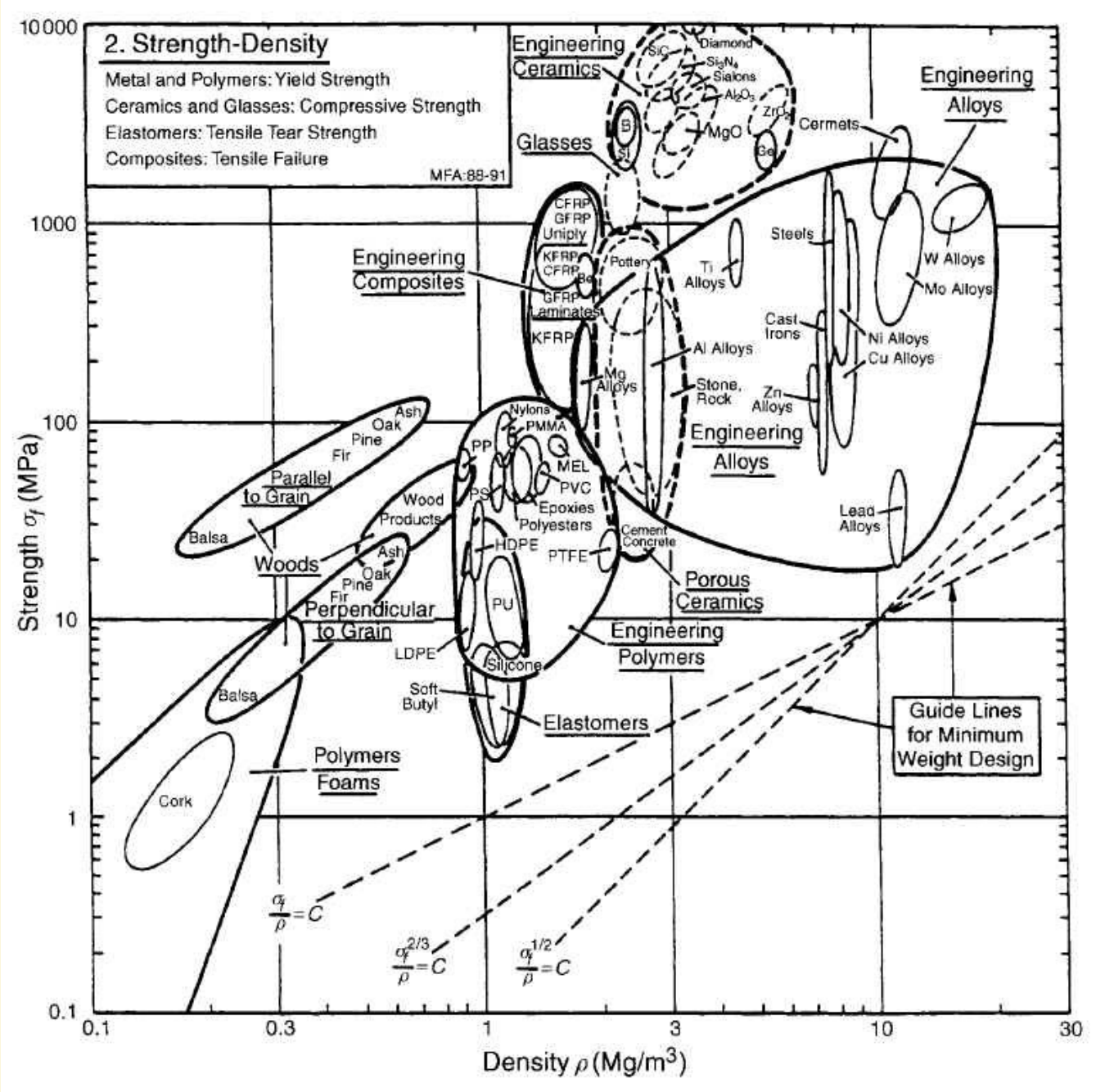
Mechanical properties

Why the differences?

- ***Lattice resistance to plastic shear***
- ***Different bond types***

Manipulating properties

- ***Metallic strengthening mechanism***
- ***Making composites***
- ***Making foams***
- ***Microstructure/processing***



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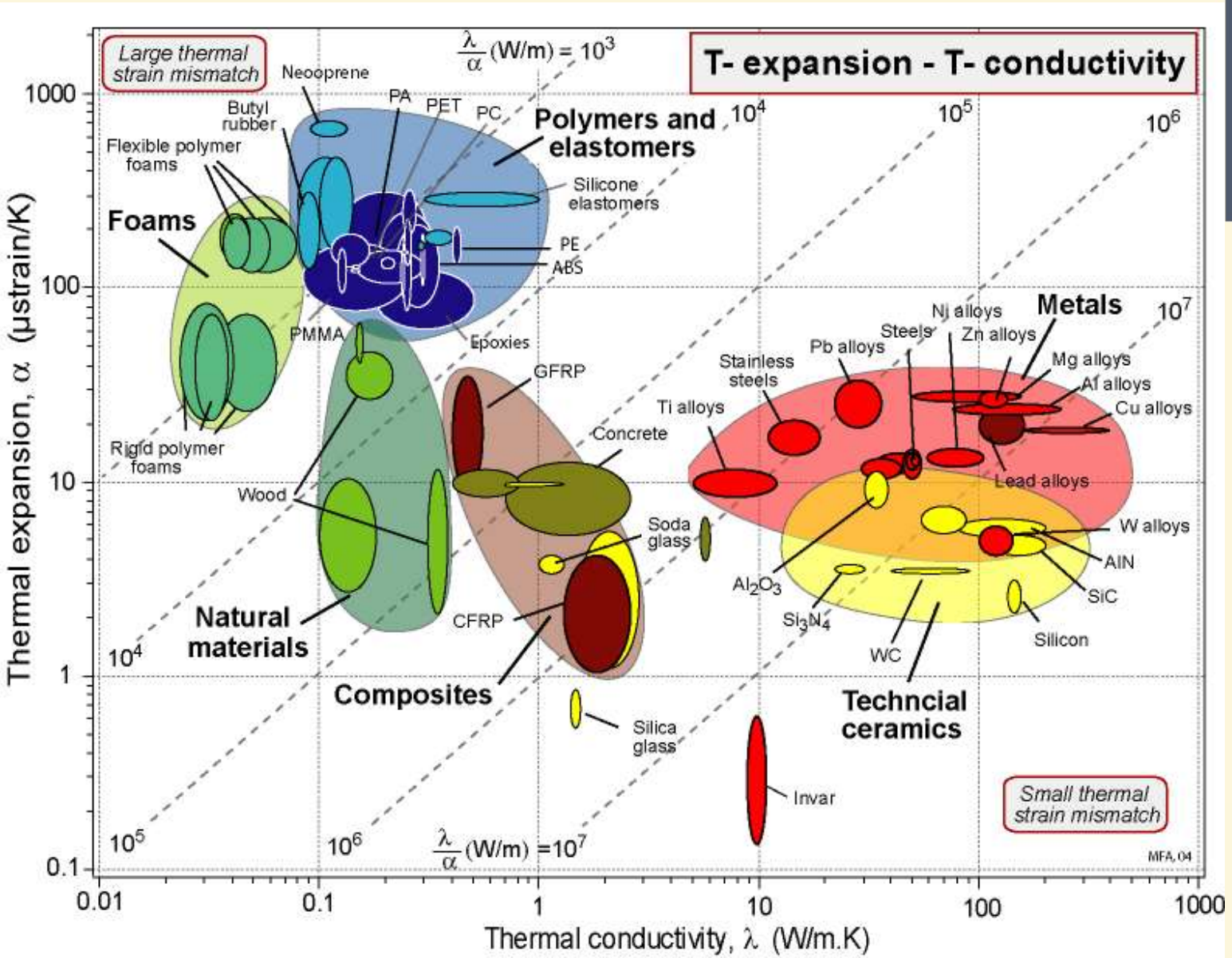
Thermal properties

Why the differences?

- *Bonds as non-linear springs*
- *10% expansion at melting point, so expansion goes inversely as T_m incr.*
- *Thermal energy as atom vibration, propagates as waves, scattered by obstacles*

Manipulating properties

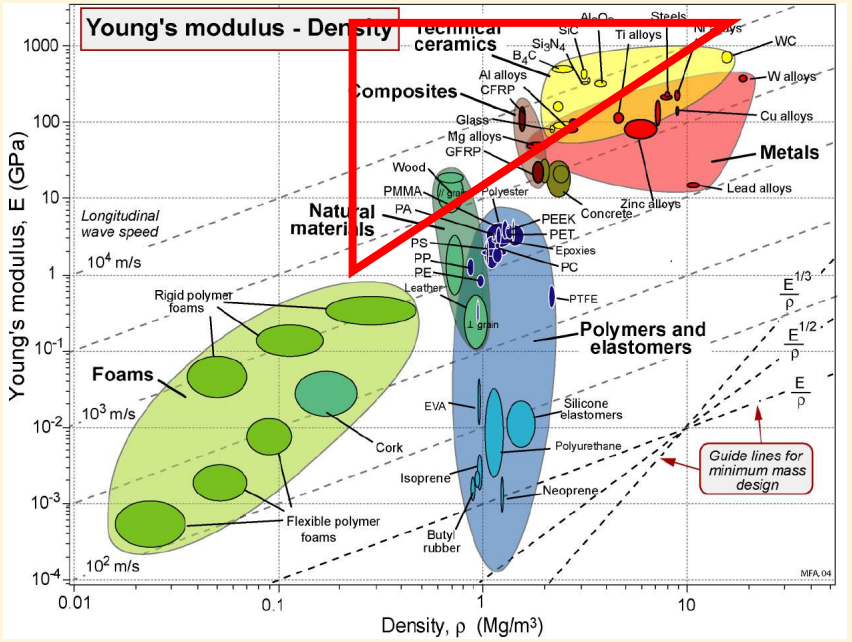
- *High conductivity: purity*
- *Low conductivity: obstacles and foams*



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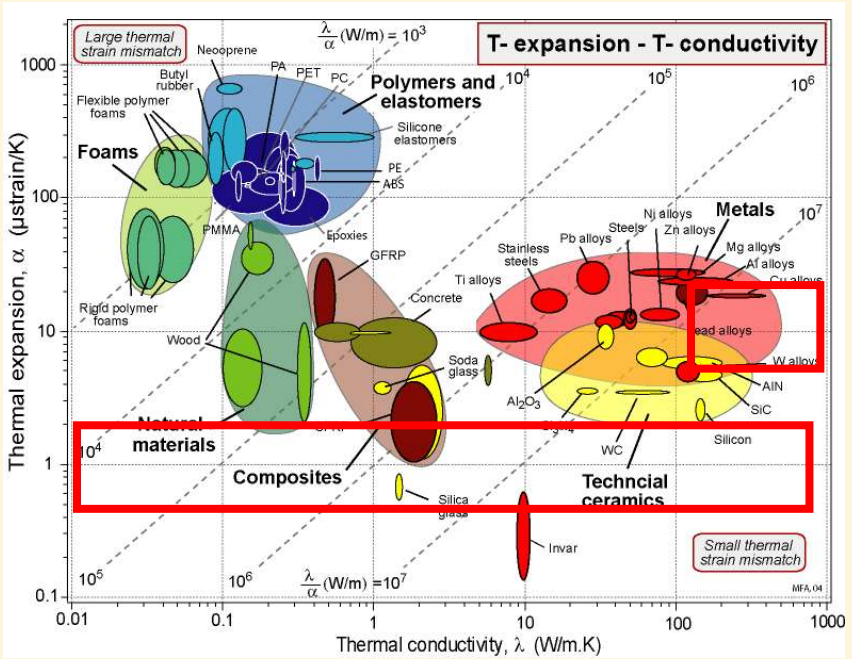
A TOOL for Materials selection

- Materials that are light and stiff?



- Materials with low expansion?

- with high conductivity?



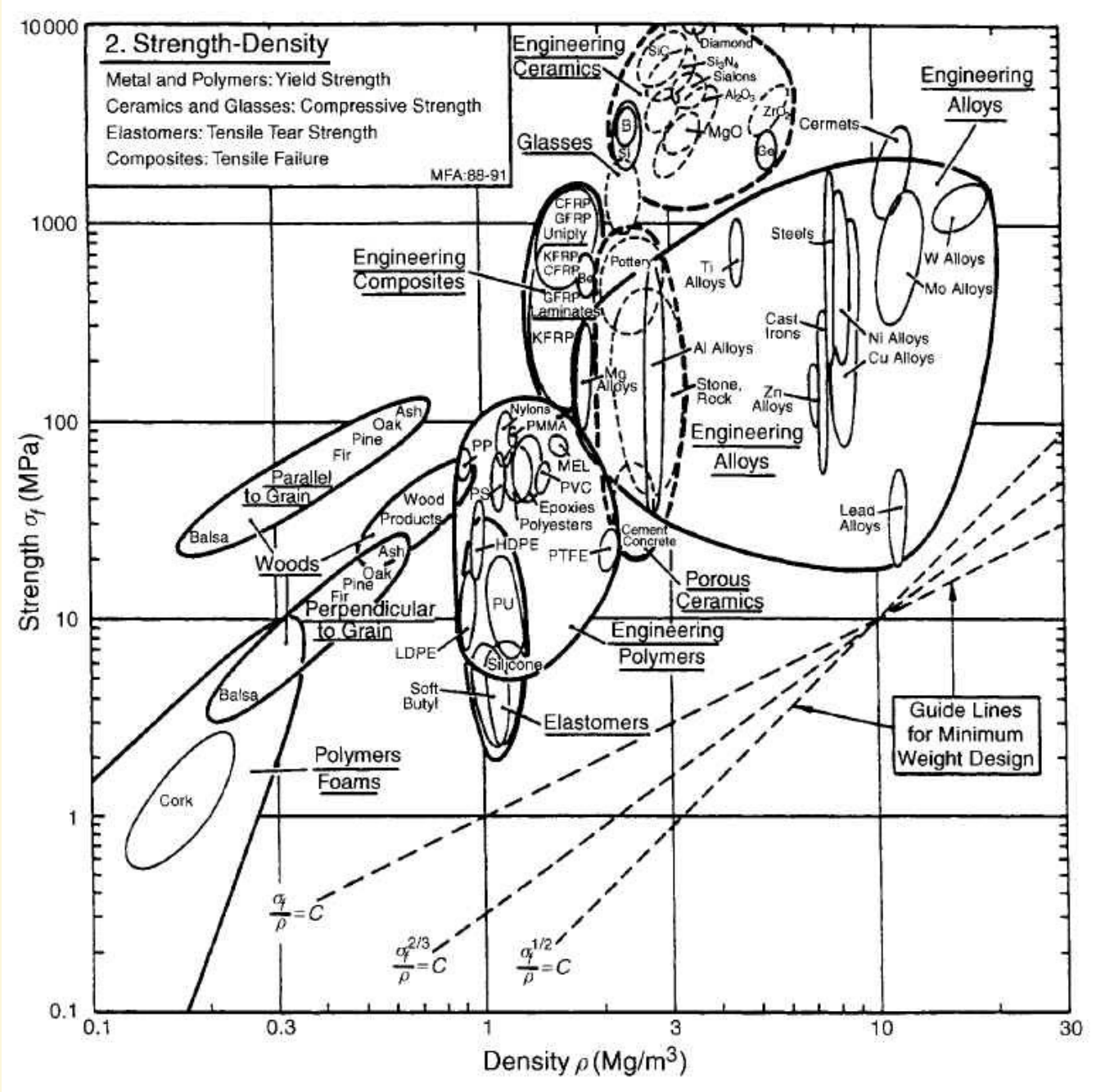
- Develops a **perspective**

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Structure Sensitivity

Is yield stress, σ_y, sensitive or insensitive to structure?

Is density, ρ, sensitive or insensitive to structure?



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Structure Sensitivity in Metals

Table 1.6 Properties of the Generic Metals

Metal	Density ($Mg\ m^{-3}$)	Young's Modulus ($GN\ m^{-2}$)	Yield Strength ($MN\ m^{-2}$)	Tensile Strength ($MN\ m^{-2}$)	Ductility	Fracture Toughness ($MN\ m^{-3/2}$)	Melting Temperature (K)	Specific Heat ($J\ kg^{-1}K^{-1}$)	Thermal Conductivity ($W\ m^{-1}K^{-1}$)	Thermal Expansion Coefficient (MK^{-1})
Iron	7.9	211	50	200	0.3	80	1809	456	78	12
Mild steel	7.9	210	220	430	0.21	140	1765	482	60	12
High-carbon steel	7.8	210	350–1600	650–2000	0.1–0.2	20–50	1570	460	40	12
Low-alloy steels	7.8	203	290–1600	420–2000	0.1–0.2	50–170	1750	460	40	12
High-alloy steels	7.8	215	170–1600	460–1700	0.1–0.5	50–170	1680	500	12–30	10–18
Cast irons	7.4	152	50–400	10–800	0–0.18	6–20	1403			
Copper	8.9	130	75	220	0.5–0.9	>100	1356	385	397	17
Brasses	8.4	105	200	350	0.5	30–100	1190		121	20
Bronzes	8.4	120	200	350	0.5	30–100	1120		85	19
Nickel	8.9	214	60	300	0.4	>100	1728	450	89	13
Monels	8.9	185	340	680	0.5	>100	1600	420	22	14
Superalloys	7.9	214	800	1300	0.2	>100	1550	450	11	12
Aluminum	2.7	71	25–125	75–135	0.1–0.5	45	933	917	240	24
1000 Series	2.7	71	28–165	75–180	0.1–0.45	45	915			24
2000 Series	2.8	71	200–500	300–600	0.1–0.25	10–50	860		180	24
5000 Series	2.7	71	40–300	120–430	0.1–0.35	30–40	890		130	22
7000 Series	2.8	71	350–600	500–670	0.1–0.17	20–70	890		150	24
Casting alloys	2.7	71	65–350	130–400	0.01–0.15	5–30	860		140	20
Titanium	4.5	120	170	240	0.25		1940	530	22	9
Ti–6 Al4V	4.4	115	800–900	900–1000	0.1–0.2	50–80	1920	610	6	8
Zinc	7.1	105		120	0.4		693	390	120	31
Lead–tin solder	9.4	40					456			
Diecasting alloy	6.7	105		280–330	0.07–0.15		650	420	110	27

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Structure Sensitivity in Metals

Structure insensitive (+/- 5%): density, elastic moduli, thermal expansion coefficient, specific heat, Poisson's ratio

Structure sensitive: yield strength, hardness, tensile strength, ductility, fracture toughness, creep strength and fatigue strength, corrosion resistance, wear resistance, thermal conductivity, electrical conductivity

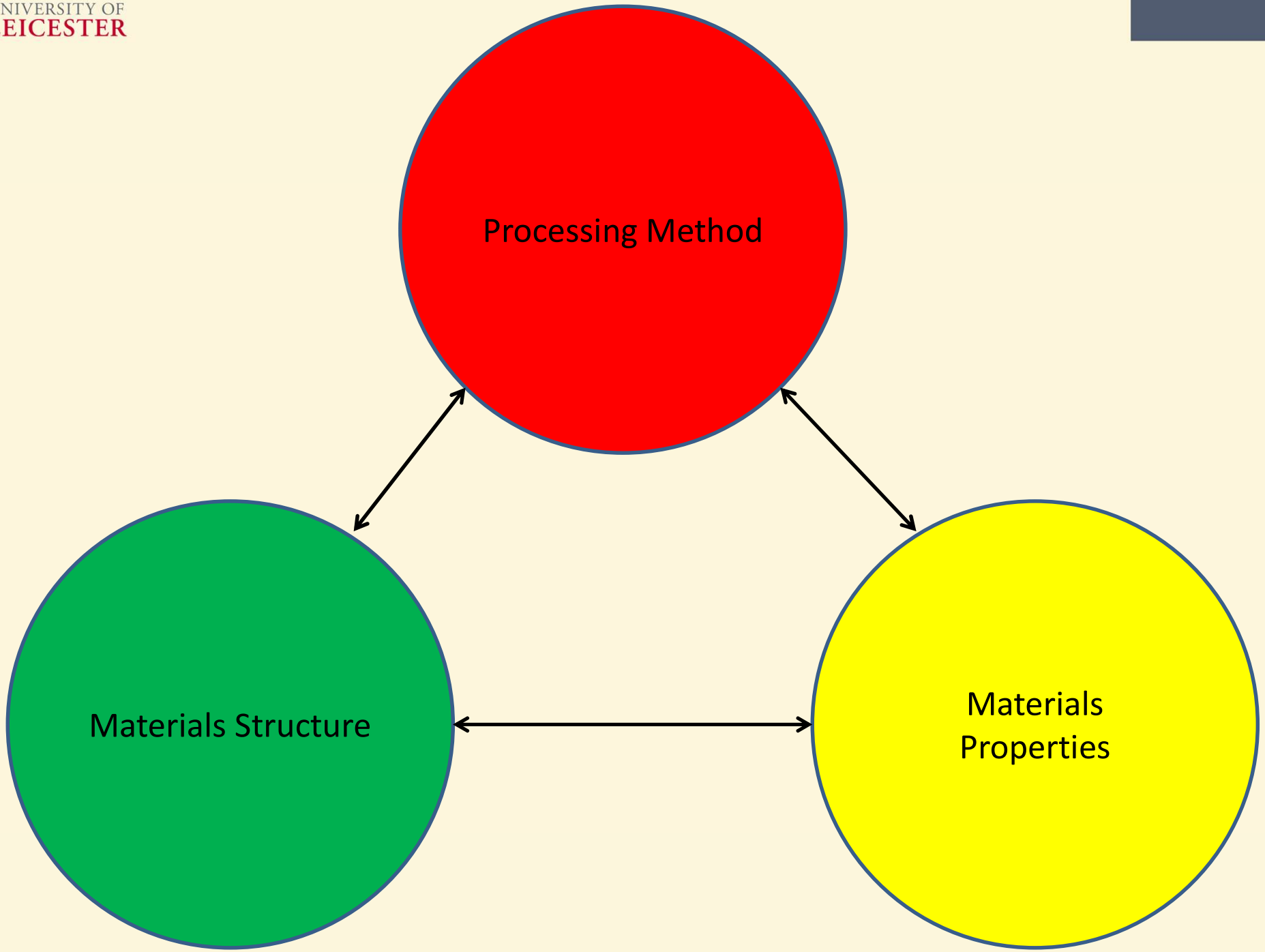
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Controlling Structure

Structure can be controlled by changing composition, processing and heat treatment

Table 1.4 Generic Aluminum-Based Metals

Metal	Typical Composition (wt%)	Typical Uses
1000 Series unalloyed Al	>99 Al	Weak but ductile and a good electrical conductor: power transmission lines, cooking foil
2000 Series major additive Cu	Al + 4 Cu + Mg, Si, Mn	Strong age-hardening alloy: aircraft skins, spars, forgings, rivets
3000 Series major additive Mn	Al + 1 Mn	Moderate strength, ductile, excellent corrosion resistance: roofing sheet, cooking pans, drinks can bodies
5000 Series major additive Mg	Al + 3 Mg 0.5 Mn	Strong work-hardening weldable plate: pressure vessels, ship superstructures
6000 Series major additives Mg + Si	Al + 0.5 Mg 0.5 Si	Moderate-strength age-hardening alloy: anodized extruded sections, e.g., window frames
7000 Series major additives Zn + Mg	Al + 6 Zn + Mg, Cu, Mn	Strong age-hardening alloy: aircraft forgings, spars, lightweight railway carriage shells
Casting alloys	Al + 11 Si	Sand and die castings
Aluminum–lithium alloys	Al + 3 Li	Low density and good strength: aircraft skins and spars

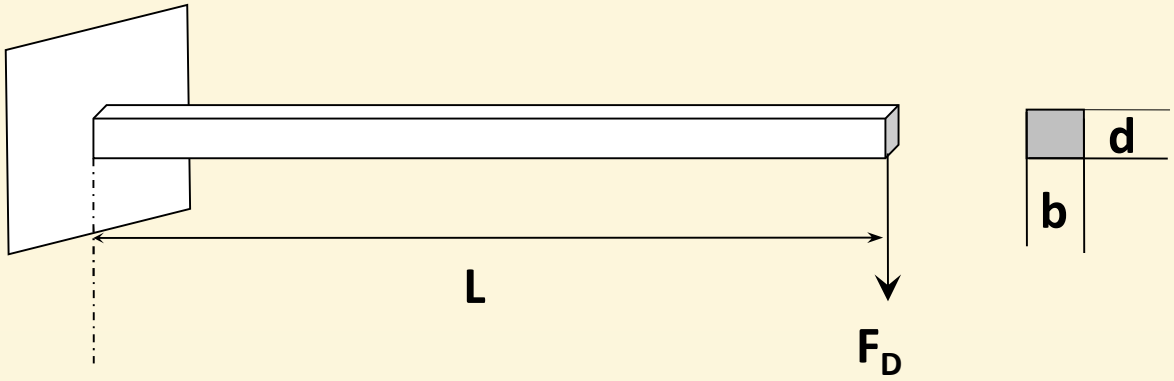

Processing Method
Materials Structure
Materials Properties

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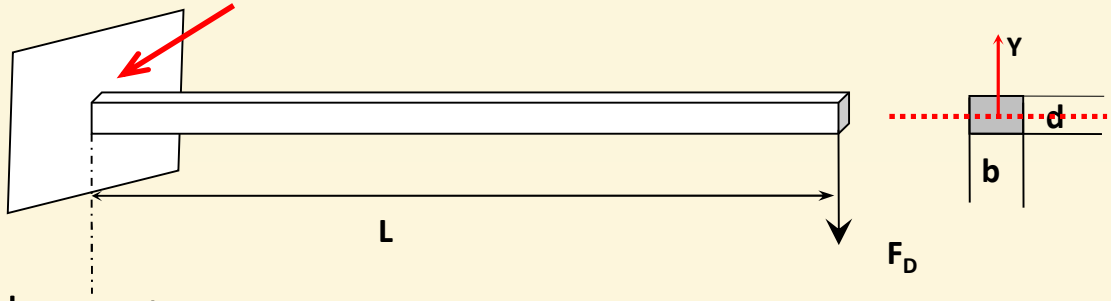
Selection of Engineering Materials

Case study 1: Design of a strong light beam:

Select a material which gives the minimum weight for a given design strength (assuming that b and L are determined by other aspects of design, but we are free to change d to carry the design force F_D without failing)



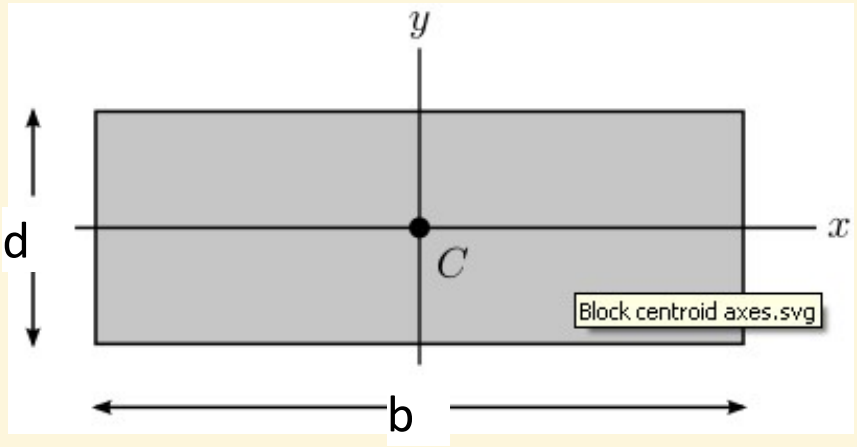
Maximum bending moment ?



Maximum bending moment is at the root:

$$M = F_D \mathrm{L} \qquad (1)$$

Maximum stress σ_{max} on the top surface of the beam at the root:



$$\sigma_{max} = \frac{My_{max}}{I} \qquad (2)$$

I – the centroidal moment of the rectangular cross section:

$$I = \frac{bd^3}{12} \qquad (3)$$

Material will fail if stress exceeds σ_f for the material. Inserting (1) and (3) into (2) gives:

$$\sigma_{\max} = \frac{My_{\max}}{I} = \frac{F_D L}{\frac{1}{12}bd^3} \times d/2 = \frac{6F_D L}{bd^2} < \sigma_f$$

We can now rearrange this to give an expression for a given material, for the minimum depth without failure, d:

$$d = \sqrt{\frac{6F_D L}{\sigma_f b}}$$

The mass of the beam is given by:

Fixed by design

Materials properties!

$$m = \rho Lbd = \rho Lb\sqrt{\frac{6F_D L}{\sigma_f b}} = Lb\sqrt{\frac{6F_D L}{b}} \times \frac{\rho}{\sqrt{\sigma_f}}$$

Fixed by design

Materials properties!

$$m = \rho L b d = \rho L b \sqrt{\frac{6F_D L}{\sigma_f b}} = L b \sqrt{\frac{6F_D L}{b}} \times \frac{\rho}{\sqrt{\sigma_f}}$$

If the remaining design parameters are fixed, then to minimise mass we need to minimise the ratio:

$$\frac{\rho}{\sqrt{\sigma_f}}$$

i.e. we seek to maximise

$$\frac{\sqrt{\sigma_f}}{\rho} = C$$

Where C is our material index

The best materials are those with the largest values of the materials index